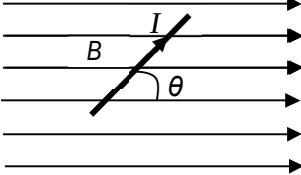
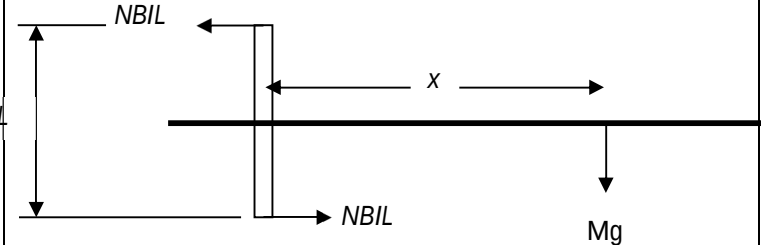


No.	Solution	Remarks
6(a)(i)	$B = \frac{F}{IL \sin \theta}$	
6(a)(ii)	 Direction of force: into plane of paper.	
6(b)(i)	From $B = \frac{F}{BIL \sin \theta}$ where $\theta = 90^\circ$ Force on opposite sides of the coil perpendicular to the field, $F = BIL \times N$  Taking moments about the axis of rotation Sum of clockwise moment = sum of anti-clockwise moment $2 (NBIL \times L/2) = Mg \times x$ $B = \frac{Mgx}{IL^2 N}$	
6(b)(ii)	- with sides of $L/2$, length of wire making the coil is halved, - resistance is halved, so current is doubled (since emf is constant) - L^2 is now $\frac{1}{4}$ of its original value, so x is halved.	
7(a)	Each metal has a characteristic minimum amount of energy.	